The mathematics first

A cylinder surface is an annulus: $(\mathbb{R}/\mathbb{Z}) \times [0,1]$. Horizontal coordinate is measured in turns, so adding an integer returns to the same place. A route is a simple arc from the fixed mark A on the bottom rim to the fixed mark B on the top, with all other points strictly between the rims. The printed tasks use upward routes. Their winding is the signed number of seam crossings. In repeated rectangles, a lift starting at $(1/2, 0)$ ends at $(1/2 + n, 1)$ exactly when the winding is the integer n.

Winding cannot change during a continuous slide on the surface with both endpoints fixed. Upward routes with the same winding can be slid into each other. A full twist is the whole-surface map

$$T_k(\theta, t) = (\theta + kt \pmod 1, t), \qquad k \in \mathbb{Z}.$$

It fixes both rims pointwise, adds k to every joining route's winding, and has inverse T_{-k}. Composition adds: $T_j T_k = T_{j+k}$. Thus a twist word's effect is its number of pluses minus minuses; its shortest equivalent word has that signed sum's absolute value as its length.

For two simple joining routes of windings a, b with *these same fixed endpoints*, the smallest number of interior intersections is

$$i(a, b) = \max\{|a - b| - 1, 0\}.$$

Exclude the shared endpoints; shared pieces and non-crossing touches are not allowed final comparisons. Arbitrary drawings may have extra intersections. A common twist preserves the minimum. Twisting only one route can increase, decrease or leave it unchanged. Children may discover a recoverable winding record, cancellation, and how a difference controls unavoidable crossings.

Who this fits. One Grades 4–5 prototype for children ready to track signed crossings, transfer a route between a cylinder and repeated strips, and keep endpoints fixed. An adult may read or record. Sliding and lift arguments need a mathematically confident adult; the general lower bound is a later extension. There is no separate younger packet or classroom-tested age-fit claim.

Materials and preparation

At the older three-child table, use a pair and rotating checker with one adult. Per pair: the four-page packet, one extra page 1 for cutting, two or three spare copies of page 2, two contrasting yarns about 60 inches long for later high windings, removable tape, scissors, pencils and blank paper. Coil excess yarn; shorter pieces suffice for initial small-winding play.

Print at actual size. The cut rectangle is 6 by 2.4 inches. Tape its left and right edges together without overlap; the dotted quarter-height lines should meet. Keep the cylinder still, with A and B vertically aligned away from the seam. Secure yarn at the geometric marks while leaving spare length beyond both ends. Rehearse feeding slack through the fastening without shifting either mark. Also rehearse whether a laid route stays put: taut yarn pulls toward the shortest, helical route, which suits Problem 6 but can undo a different-looking route in Problem 1 or the middle of a slide in Problem 3. Let children hold the yarn with fingers or add small pieces of tape along it. If tape grips too tightly or the sleeve buckles, improve the fastening or keep a drawn route as the mathematical record. This physical operation is unrehearsed.

Launch and a feasible first visit

Let children handle the tube and yarn. Trace the printed seam example: an exit through the right edge re-enters the left edge at the *same height*, counting $+1$. Show a route with fixed A and B and let a partner check it. Demonstrate how one continues into the next repeated rectangle before giving out page 2. Spend the first visit on Problems 1–3: route building, recording and same-winding sliding may use the full hour. A useful stopping point is two different-looking routes sharing a winding and an explanation of why different windings cannot slide together. Save twist words and intersection minimization for a return visit or a pair already secure with the cylinder-to-strip bridge.

Answers and optional hints

Problem 1

All four requested windings exist. In normalized repeated rectangles, draw the straight lift $x(t) = 1/2 + nt$, $0 \le t \le 1$, for $n = 0, 1, -1, 2$. Project the route back to the cylinder by wrapping x modulo 1. These are simple upward routes with signed seam totals $0, +1, -1, +2$. To change appearance without changing winding, add a small sideward bulge away from both endpoints. For example, $x(t) = 1/2 + nt + \frac{1}{4}t(1-t)$ has the same endpoints.

Hint: count signed crossings as you travel from A to B, rather than counting every contact with the seam as a positive turn. Move a route slightly if it lies along the seam or merely touches it; the counting convention needs clear transverse crossings. A drawing or recorded lift is more reliable than remembering how the yarn was laid.

Problem 2

Winding $+2$ ends at B in copy 2; winding -1 at B in copy -1. With A at $(1/2, 0)$, the endpoints are $(5/2, 1)$ and $(-1/2, 1)$. For each winding, the straight and gently bulged lifts above give two examples. Every passage through a vertical copy boundary corresponds to a seam crossing; height stays unchanged when returning to the base rectangle. The copy endpoint records the net winding even for routes with extra opposite seam crossings. *Hint:* ask the partner to point to the matching B copy before smoothing a drawn route.

Problem 3

Two distinct routes with the same winding can slide together; a pair with different windings cannot. Use the straight and bulged winding-$+1$ routes as the first pair, and windings $0, +1$ as the second. For upward lifts $x_0(t), x_1(t)$ with the same endpoint copy, the interpolation $(1-s)x_0(t) + sx_1(t)$, $0 \le s \le 1$, is an explicit slide. Height t remains unique on each route, so no route crosses itself; endpoints remain fixed.

During any continuous fixed-endpoint slide, the lifted top endpoint moves continuously but must always be one of the separated B copies. It cannot jump from one copy to another. This proves winding invariance and the impossibility for different windings. A different-looking route is not necessarily a different winding. *Hint:* "Which B copy would its endpoint have to reach?"

Problem 4

Words have the same effect exactly when their signed sums agree. From winding 0, a word with p pluses and m minuses ends at winding $p - m$. Its shortest replacement is $p - m$ pluses if positive, $m - p$ minuses if negative, and the empty word if zero. Each twist changes winding by only 1, so fewer than $|p - m|$ twists cannot suffice; repeating the needed sign attains it. For examples, $++-$ has winding 1 and reduces to $+$; $+--$ reduces to $-$; $+--+$ reduces to no twists. The strip under Problem 4 prints B copies -4 to 4: each of the 28 words of two to four twists ends there, and so does every route along the way.

The displayed shifts are fractions of a full turn: at heights $0, 1/4, 1/2, 3/4, 1$, a plus twist shifts by those same turn amounts. At the top a full turn fixes every point modulo 1. Adding shifts proves $T_j T_k = T_{j+k}$ on the whole surface, so cancellation works on every route. Do not physically rotate a marked rim to imitate this map. When page 3 starts, demonstrate one $+$ twist on a copy of the strip under Problem 4, using a winding-0 route with one bend: move the bend and the points on the dotted lines rightward by their height in copies, move the top end one whole copy, and join the moved points with straight pieces (a twist keeps straight pieces straight). Then re-lay the yarn to match.

Problem 5

Any word with three pluses and three minuses works, such as $+++--$. It has net shift zero, so composition is T_0, the identity, regardless of the starting winding. Conversely, a length-six word has net zero only if it contains three of each sign. There are $\binom{6}{3} = 20$ such words if children choose to catalog them. Trying each possible starting route is unnecessary because the same signed sum is added to all of them.

Problem 6

Windings	$(0,1)$	$(0,2)$	$(0,3)$	$(-1,1)$	$(-1,2)$	$(2,4)$
Fewest strand crossings	0	1	2	1	2	1

The page counts places where the two strands cross each other, away from A and B. Seam crossings are not counted here; counting them instead would give $|a| + |b|$.

Construction. For unequal windings use straight lifts $(1/2+at, t)$ and $(1/2+bt, t)$. Their projections meet at an interior height exactly when $(a-b)t$ is an integer. For $d = |a-b| \geq 1$, those heights are $1/d, 2/d, \ldots, (d-1)/d$, giving $d-1$ crossings. They are distinct interior points and genuine crossings. When $a = b$, put one route slightly alongside the other, with the sideward gap shrinking to zero only at A and B. There are then zero interior crossings; laying identical yarns on top of each other is not a valid zero-crossing answer.

Why no drawing can do better. For upward lifts, the horizontal difference of the two routes varies continuously from 0 to $b-a$. It must attain each integer strictly between those values. At every such height the projected routes meet, and different attained integers need different heights. Thus at least $|a-b|-1$ interior intersections are forced if the difference has absolute value at least 2. For differences 0 or 1, zero is the lower bound.

The same bound holds even for proper simple arcs that wiggle up and down. Cut along the first arc to form a rectangle, straighten its two cut edges, then reglue. This is a boundary-fixing homeomorphism sending it to a vertical arc; the other arc's relative winding becomes $b-a$. In repeated copies, the first arc's lifts are integer-spaced vertical barriers. A lift of the second must cross each barrier strictly between its starting and ending copies. Simplicity prevents two such forced meetings from being the same projected interior point. Together with the construction, this proves the minimum rather than just a pattern in examples. This general cut-and-reglue argument can remain with the adult.

Hint: compare the two top copies in a repeated strip, or first test one nearly straight vertical route against a winding route. Do not announce the difference rule before children have minimized several contrasting pairs.

Problem 7

The minima are 4 for $(2,7)$ and 0 for $(5,5)$. The general rule is $\max(|a-b|-1, 0)$. Adding the same integer twist k replaces (a,b) by $(a+k, b+k)$, leaving the difference and the minimum unchanged. A twist on only one route can increase, decrease or preserve the answer: $(0,2) \to (0,3)$ changes $1 \to 2$; $(0,2) \to (1,2)$ changes $1 \to 0$; $(5,5) \to (6,5)$ stays $0 \to 0$.

Physical meaning and source limits

The rigid tube is a stationary strand-routing model. Lifting yarn to construct a new route is permitted; it is not a legal continuous surface slide and cannot prove two surface routes disjoint. Visible yarn overlaps represent intersections of their projected centerlines on the surface, not knot over/under information. No literal twisting paper sleeve is required. Keep fixed endpoints, rim marks, proper interiors and excluded shared endpoints throughout; changing them changes the question and can change the formula.

Diana Davis, *Billiards, Surfaces, and Geometry*, Chapter 4, Problem 111 and Figure 83, printed p. 111 (PDF p. 129), supplies related torus-shear and reassembly imagery. It does not give this fixed-endpoint annulus model or intersection formula; those are independently developed and proved here.

Review status: matched to the four-page GGT75-S-v2 student pages. Version 2 (October 23, 2026) names strand crossings in Problems 6–7, adds the twist strip under Problem 4 and taller sketch rows in Problem 6; problem numbers are unchanged. The independent checker covers the exact printed instances, twist words and rational composition tests; general results use the proofs above. The guide is built, rendered and visually checked. Physical preparation, surface sliding, yarn fit and classroom pacing remain unrehearsed and unpiloted.